

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/4 & 3/8 & 1/4 \\ 0 & -1/2 & 1/2 \\ 0 & 0 & 1/2 \end{bmatrix} \quad U^{-1} = \begin{bmatrix} 1/4 & 3/8 & 1/4 \\ 0 & -1/2 & 1/2 \\ 0 & 0 & 1/2 \end{bmatrix}$$

$$[L|I] \rightarrow [I|L^{-1}]$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{array}{l} \text{Row 2} \\ = \text{Row 2} \\ + \text{Row 1} \end{array}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{array}{l} \text{Row 3} = \text{Row 3} \\ -2 \times \text{Row 1} \end{array}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = U^{-1}L^{-1} = \begin{bmatrix} 1/8 & 3/8 & 1/4 \\ -3/2 & -1/2 & 1/2 \\ -1 & 0 & 1/2 \end{bmatrix} \quad \square$$

2) Suppose  $A=BC$ , where  $B$  is invertible. Show that any sequence of row operations that reduces  $B$  to  $I$  also reduces  $A$  to  $C$ . The converse is not true, since the zero matrix may be factored as  $0=B \cdot 0$ .

Solution: By Theorem 7, since  $B$  is invertible,  $B$  is row equivalent to  $I$ .

$\therefore$  There exists Elementary matrices  $E_1, E_2, \dots, E_p$  s.t.

$(E_p E_{p-1} \dots E_2 E_1) B = I$ . By Defn. of inverse, there exists a matrix  $B^{-1}$  of the same size as  $B$  s.t.  $B^{-1}B = BB^{-1} = I$ .

$$((E_p E_{p-1} \dots E_2 E_1) B) B^{-1} = I B^{-1}$$

$$\Rightarrow (E_p E_{p-1} \dots E_2 E_1) (BB^{-1}) = I B^{-1} \quad (\text{Theorem 2(a)})$$

$$\Rightarrow (E_p E_{p-1} \dots E_2 E_1) I = I B^{-1} \quad (\text{Defn. of inverse})$$

$$\Rightarrow B^{-1} = E_p E_{p-1} \dots E_2 E_1 \quad (\text{Theorem 2(e)})$$

$$A=BC \Rightarrow B^{-1}A = B^{-1}(BC) \Rightarrow B^{-1}A = (B^{-1}B)C \quad (\text{Theorem 2(a)})$$

$$\Rightarrow B^{-1}A = IC \quad (\text{Defn. of inverse}) \Rightarrow B^{-1}A = C \quad (\text{Theorem 2(e)})$$

$$\Rightarrow (E_p E_{p-1} \dots E_2 E_1) A = C.$$

$\therefore$  The sequence of row operations that reduces  $B$  to  $I$  also reduces  $A$  to  $C$ .

$\square$